

EXPERIMENT - 2

```
[mysql> USE bookflow_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
```

Database changed

```
[mysql> SHOW TABLES;
```

```
+-----+
| Tables_in_bookflow_db |
+-----+
| Books                  |
| Loans                  |
| Members                |
+-----+
```

3 rows in set (0.001 sec)

```
mysql> CREATE TABLE Donation_History (
->   donation_id INT PRIMARY KEY,
->   book_id INT,
->   category VARCHAR(50),
->   donation_date DATE,
->   FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
```

Query OK, 0 rows affected (0.017 sec)

```
[mysql> SHOW TABLES;
```

```
+-----+
| Tables_in_bookflow_db |
+-----+
| Books                  |
| Donation_History       |
| Loans                  |
| Members                |
+-----+
```

4 rows in set (0.001 sec)

```
[mysql> DESCRIBE Donation_History;
```

Field	Type	Null	Key	Default	Extra
donation_id	int	NO	PRI	NULL	
book_id	int	YES	MUL	NULL	
category	varchar(50)	YES		NULL	
donation_date	date	YES		NULL	

4 rows in set (0.005 sec)

```
[mysql> SELECT * FROM Members;
```

member_id	full_name	email
101	Amrutha	amrutha@gmail.com
102	Rahul	rahul@gmail.com
103	Priya	priya@gmail.com

3 rows in set (0.001 sec)

```
mysql> SHOW INDEX FROM Books;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
Books	0	PRIMARY	1	book_id	A	3	NULL	NULL		BTREE			YES	NULL
Books	0	isbn	1	isbn	A	3	NULL	NULL	YES	BTREE			YES	NULL

2 rows in set (0.007 sec)

```
mysql> CREATE INDEX idx_isbn
```

```
-> ON Books(isbn);
```

Query OK, 0 rows affected (0.027 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> SHOW INDEX FROM Books;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Table | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Books | 0 | PRIMARY | 1 | book_id | A | 3 | NULL | NULL | YES | BTREE | | | YES | NULL |
| Books | 0 | isbn | 1 | isbn | A | 3 | NULL | NULL | YES | BTREE | | | YES | NULL |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.007 sec)

mysql> CREATE INDEX idx_isbn
-> ON Books(isbn);
Query OK, 0 rows affected (0.027 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SHOW INDEX FROM Books;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Table | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Books | 0 | PRIMARY | 1 | book_id | A | 3 | NULL | NULL | YES | BTREE | | | YES | NULL |
| Books | 0 | isbn | 1 | isbn | A | 3 | NULL | NULL | YES | BTREE | | | YES | NULL |
| Books | 1 | idx_isbn | 1 | isbn | A | 4 | NULL | NULL | YES | BTREE | | | YES | NULL |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.008 sec)

mysql> SELECT * FROM Donation_History;
+-----+-----+-----+-----+
| donation_id | book_id | category | donation_date |
+-----+-----+-----+-----+
| 1 | 4 | Programming | 2026-08-03 |
+-----+-----+-----+-----+
1 row in set (0.005 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.000 sec)
```